

Adversarial Quiz Explanations

Google Form Link: <https://forms.gle/psxVpkW1Uvk92htNA>

This document explains why each correct answer is correct and why each incorrect option is a plausible misconception or edge-case trap.

Q1. What does a convolution kernel primarily do?

- A. Sort pixels
- B. Multiply and sum over a region 
- C. Normalize image
- D. Compress image

Why B is Correct

A convolution kernel slides across an image region, multiplying kernel values with pixel values and summing the results to extract features.

Why the Other Options Are Tempting

- A. Sort pixels
Sounds computational, but convolutions never rearrange pixel order.
 - C. Normalize image
Normalization is a preprocessing step, not the kernel's main role.
 - D. Compress image
Convolution may reduce dimensions indirectly, but its primary purpose is feature extraction.
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Q2. What is the output size if:

$N = 32, F = 3, P = 1, S = 1$

- A. 30
- B. 31
- C. 32 
- D. 33

Formula

$$\text{Output} = ((N - F + 2P) / S) + 1$$

$$\text{Output} = ((32 - 3 + 2(1)) / 1) + 1$$

$$\text{Output} = (31 / 1) + 1$$

$$\text{Output} = 32$$

Why C is Correct

Padding preserves the original spatial dimensions when using a 3×3 filter and stride 1.

Why the Other Options Are Tempting

- A. 30
Happens when padding is ignored.
 - B. 31
Common mistake from forgetting the final "+1".
 - D. 33
Caused by adding padding incorrectly.
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Q3. Increasing stride results in:

- A. Larger output
- B. Smaller output 
- C. More padding
- D. More channels

Why B is Correct

A larger stride skips more positions while sliding the kernel, reducing output dimensions.

Why the Other Options Are Tempting

- A. Larger output
Students sometimes assume "bigger stride" means "bigger result."
 - C. More padding
Padding and stride are independent parameters.
 - D. More channels
Channels are determined by filter count, not stride.
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Q4. Which filter is best for noise reduction?

- A. Sobel
- B. Laplacian
- C. Gaussian 
- D. Max pooling

Why C is Correct

Gaussian filters smooth images and suppress high-frequency noise.

Why the Other Options Are Tempting

- A. Sobel
Sobel detects edges rather than reducing noise.
 - B. Laplacian
Laplacian often amplifies noise.
 - D. Max pooling
Pooling reduces dimensions but is not a smoothing filter.
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Q5. Sobel filter is used for:

- A. Blurring
- B. Edge detection 
- C. Compression
- D. Color correction

Why B is Correct

Sobel filters compute image gradients to highlight edges.

Why the Other Options Are Tempting

- A. Blurring
Gaussian filtering performs blurring.
 - C. Compression
Compression is unrelated to edge filtering.
 - D. Color correction
Sobel works on gradients, not color balancing.
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Q6. Padding is used to:

- A. Increase computation
- B. Preserve spatial size 
- C. Reduce channels
- D. Normalize input

Why B is Correct

Padding helps maintain output dimensions and preserves edge information.

Why the Other Options Are Tempting

- A. Increase computation
Padding can slightly increase operations, but that is not its purpose.
- C. Reduce channels
Channels depend on filters.

- D. Normalize input
Normalization is separate from padding.
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Q7. Depthwise convolution applies:

- A. One filter per channel 
- B. One filter to all channels
- C. No filters
- D. Random filters

Why A is Correct

Depthwise convolution processes each channel independently with its own filter.

Why the Other Options Are Tempting

- B. One filter to all channels
This resembles standard convolution.
 - C. No filters
Convolution always uses filters.
 - D. Random filters
Filters are learned during training.
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Q8. Pointwise convolution uses:

- A. 3×3 filters
- B. 5×5 filters
- C. 1×1 filters 
- D. No filters

Why C is Correct

Pointwise convolution uses 1×1 filters to combine information across channels.

Why the Other Options Are Tempting

- A. 3×3 filters
Common convolution size but not pointwise convolution.
 - B. 5×5 filters
Larger spatial filter, not pointwise.
 - D. No filters
Pointwise convolution still uses learnable kernels.
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Q9. Laplacian filter detects:

- A. Color
- B. Texture
- C. Noise
- D. Second-order edges 

Why D is Correct

The Laplacian operator detects rapid intensity changes using second derivatives.

Why the Other Options Are Tempting

- A. Color
Laplacian works on intensity changes, not color.
 - B. Texture
Texture analysis is broader and indirect.
 - C. Noise
Laplacian may amplify noise but does not specifically detect it.
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Q10. Which parameter increases computation most?

- A. Padding
- B. Kernel size 
- C. Stride
- D. Activation

Why B is Correct

Larger kernels require significantly more multiplications and parameters.

Why the Other Options Are Tempting

- A. Padding
Padding changes dimensions slightly but has smaller impact.
- C. Stride
Larger stride usually reduces computation.
- D. Activation
Activations add computation but far less than large kernels.